

Phylogenetic Imputation of Trait Data: Methods, Benchmarks, and Best Practices

A Literature Review Generated with AI Assistance

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Identifier: 2608.00005

2026-08-25

Abstract

Missing trait values are a pervasive obstacle in comparative biology and macroecology, motivating a rich literature on phylogenetic imputation. This review benchmarks the principal families of methods: Bayesian phylogenetic mixed models, multivariate regression trees, missForest with phylogenetic predictors, and Rphylopars-style variance-covariance approaches. Using published simulation criteria — error magnitude, bias, coverage of credible intervals, and sensitivity to phylogenetic uncertainty — we compare performance across trait types (continuous vs. categorical), missingness regimes, and tree sizes. We distill practical guidance: report imputation uncertainty in downstream analyses, prefer multivariate over univariate imputation when correlated traits exist, and validate on held-out taxa rather than random entries.

Keywords: phylogenetic imputation, trait evolution, comparative methods, missing data.

Placeholder Notice

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Scope

The forthcoming review will synthesize recent literature on the topic above, covering its principal methods, findings, open challenges, and directions for future work within the field of Ecology & Evolution.

Data and Code Availability

No new data or code are associated with this placeholder.

References

- [1] LitReview Consortium (2026). *LitReview: an arXiv-style repository of AI-generated literature reviews for the basic sciences*. ai.litreview.org.